

would be difficult to generalize their behavior. It would also be difficult to project their behavior in time.

In orthodox economics, rationality became a straitjacket. Platonified economists ignored the fact that people might prefer to do something other than maximize their economic interests. This led to mathematical techniques such as “maximization,” or “optimization,” on which Paul Samuelson built much of his work. Optimization consists in finding the mathematically optimal policy that an economic agent could pursue. For instance, what is the “optimal” quantity you should allocate to stocks? It involves complicated mathematics and thus raises a barrier to entry by non-mathematically trained scholars. I would not be the first to say that this optimization set back social science by reducing it from the intellectual and reflective discipline that it was becoming to an attempt at an “exact science.” By “exact science,” I mean a second-rate engineering problem for those who want to pretend that they are in the physics department—so-called physics envy. In other words, an intellectual fraud.

Optimization is a case of sterile modeling that we will discuss further in [Chapter 17](#). It had no practical (or even theoretical) use, and so it became principally a competition for academic positions, a way to make people compete with mathematical muscle. It kept Platonified economists out of the bars, solving equations at night. The tragedy is that Paul Samuelson, a quick mind, is said to be one of the most intelligent scholars of his generation. This was clearly a case of very badly invested intelligence. Characteristically, Samuelson intimidated those who questioned his techniques with the statement “Those who can, do science, others do methodology.” If you knew math, you could “do science.” This is reminiscent of psychoanalysts who silence their critics by accusing them of having trouble with their fathers. Alas, it turns out that it was Samuelson and most of his followers who did not *know* much math, or did not know how to use what math they knew, how to apply it to reality. They only knew enough math to be blinded by it.

Tragically, before the proliferation of empirically blind idiot savants, interesting work had been begun by true thinkers, the likes of J. M. Keynes, Friedrich Hayek, and the great Benoît Mandelbrot, all of whom were displaced because they moved economics away from the precision of second-rate physics. Very sad. One great underestimated thinker is G.L.S. Shackle, now almost completely obscure, who introduced the notion of “unknowledge,” that is, the unread books in Umberto Eco’s library. It is unusual to see Shackle’s work mentioned at all, and I had to buy his books from secondhand dealers in London.

Legions of empirical psychologists of the heuristics and biases school have shown that the model of rational behavior under uncertainty is not just grossly